46TH ANNIVERSARY

THE NEXT 100 INNOVATORS

SWARM SAPIEN

TANMAY BUNKAR, 30

Founder, Botlab Dynamics Delhi

"Finance is the biggest constraint in technology, not physics"

When Tanmay Bunkar, a young IIT Delhi student, read of how the Indian army was looking to import American-made Javelin anti-tank missiles in 2010, he began looking for solutions. He quickly gave up because the sugar rockets—homemade rockets using potassium nitrate—were proving expensive to build. Less than a decade later, he had found the answer—dozens of low-cost drones carrying explosives and flying in a 'swarm' could attack tanks far beyond the range of anti-tank missiles. By 2014, he had demonstrated a two-drone hunter-killer team. In 2016, he began Botlab Dynamics, a start-up at IIT Delhi, with a loan of Rs 50 lakh from the Department of Science and Technology.

He didn't win the 2019 IAF competition to field drones but used the experience to branch out into swarm drones where packs of armed drones fly together like herds of killer bees, in perfect symmetry and saturate enemy air defences or attack soft-skinned vehicles, aircraft and troops in the open. Bunkar is now one of a handful of Indian start-ups offering completely homegrown products. The breakthrough came last year when Bunkar, the son of a decorated policeman from Morena in Madhya Pradesh, perfected his own algorithm-based flight controller—which tells the machine how to fly. His firm takes on civilian contracts for mass drone displays where pre-programmed drones make patterns in the sky. This isn't his desired end-state for his technology, but Bunkar sees it essential for revenues. ■

–Sandeep Unnithan

ROLE MODEL His father, a decorated Senior Inspector of Police in Madhya Pradesh, passed away due to pandemic-related stress in 2020

SANDEEP SINGH